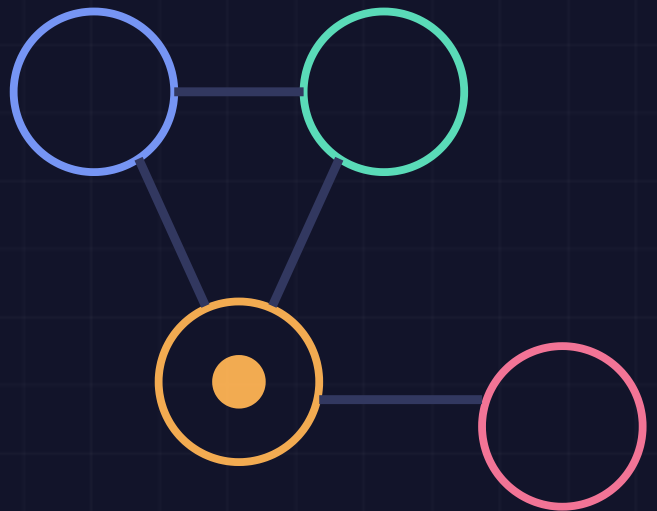


Neuron.

The rules behind it

How a lesson is scored, what opens next, how levels and jigsaw pieces are earned, what the grown-ups' settings change, and why a tampered save file cannot cheat any of it.



Neuron · repository [miekuskamil/ai-academy-game](#)

Build reviewed: branch [review-fixes](#), commit [474e92a](#)

26 September 2026 · document 3 of 3 ([Overview](#) · [Architecture](#) · [Logic](#))

Store what happened, work out the rest

The save file keeps a record per lesson of what she did. Everything else — which lessons are open, her level, which jigsaw pieces she has earned, whether the picture is complete — is recomputed from those records every time the screen draws.

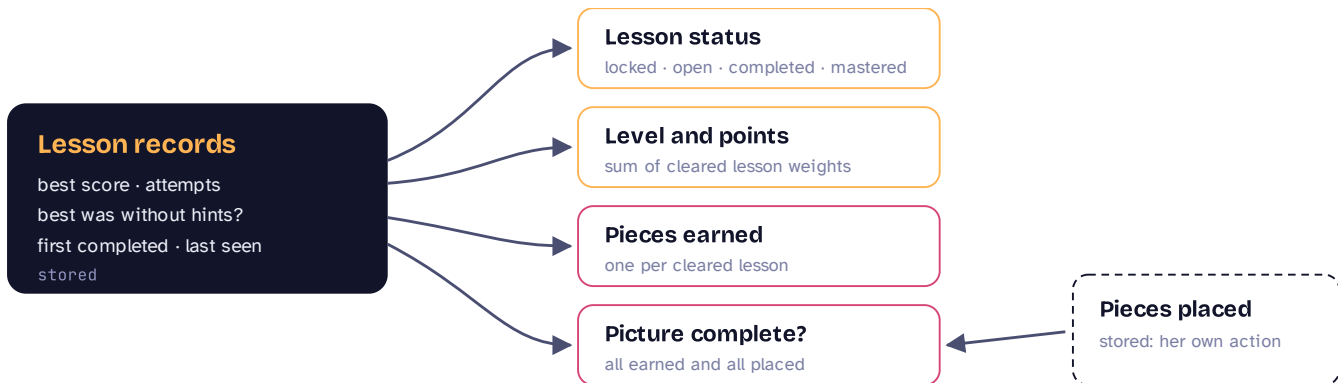


Figure 1. Only the dark box and the dashed box are stored. Every other value is derived on each read, so there is no second copy to fall out of step.

Why this matters. A save file edited by hand cannot inflate her level or open the picture: those values are never read from the file. Import and export also cannot drift out of sync, because there is only one source.

Recording an attempt

When she presses “See how I did” and the lesson passes, `ProgressService.record()` updates that lesson's record:

```

best = max(previous best, this score) // a retry can only raise it
attempts = previous attempts + 1
bestUnaided = if this attempt is the new best: no hints were used
               otherwise: keep the old value // follows the best attempt
completedAt = first time best reached 0.6; never cleared later
  
```

It then saves, re-derives everything, and announces what changed: `lesson:completed`, `lesson:mastered`, `lesson:unlocked`, `level:gained` and always `progress:changed`.

Status and what opens next

STATUS	RULE
mastered	Best score ≥ 0.9 and that best attempt used no hints.
completed	Best score ≥ 0.6 . This is “cleared”: it earns the piece and the points, and unlocks what follows.
open	Not yet cleared, and every prerequisite is met.
locked	Anything else. Opening a locked lesson by its address shows “Not open yet”.

Each lesson lists its prerequisites in its content file. In this course they form one straight line: every lesson needs the one before it, starting from “Ask like you mean it”.

Hidden worlds

A grown-up can hide a world. A prerequisite inside a hidden world does not block; it counts as met once its own prerequisites are met. So the chain skips over the hidden world instead of stopping at it.



The Lesson tab and “Carry on” button skip hidden lessons; opening one directly shows “Put away for now”.

Hiding never deletes records. Showing the world again brings back its lessons exactly as they were. The picture still needs a piece from every lesson, so the Puzzle page says how many pieces belong to worlds that are put away.

Levels

Each lesson has a weight equal to its depth in the chain: the first lesson is worth 1, the second 2, and so on up to 20, for 210 points in total. Her points are the sum of the weights of cleared lessons; her level is the highest threshold reached.

```

thresholds = 0, 8, 20, 35, 52, 70, 90, 111, 133, 155, 179, 204, 229, ...
// 210 points reaches level 12. Later lessons move the counter further,
// so a level reflects how far she has come, not how many she ticked off.
  
```

How answers are marked

Each question type has its own grader. Every grader returns a fraction from 0 to 1, whether it is fully right, and feedback. A lesson's score is its points earned divided by points available.

QUESTION TYPE	SCORING RULE	PARTIAL CREDIT
Pick answers one or several	Right picks divided by (right answers U her picks). Picking extra wrong answers lowers the score.	Yes
Put in order	Share of item pairs she placed in the right relative order.	Yes
Match pairs	Correct pairs divided by the number of pairs.	Yes
Give a number	Full marks within the tolerance; half marks within three times the tolerance; feedback says which way to move.	Half
Write a request	Checks each “must include” group (any phrase in the group counts) plus a minimum length. Each banned phrase costs one check, floored at zero.	Yes
Activity	The activity's own check reads what she did (see section 05).	Yes

When a lesson passes

passes = every question fully right **OR** lesson score ≥ 0.6
 // otherwise the answers stay on screen with feedback, and she fixes them
 mastered on this attempt = score ≥ 0.9 **AND** no hints opened

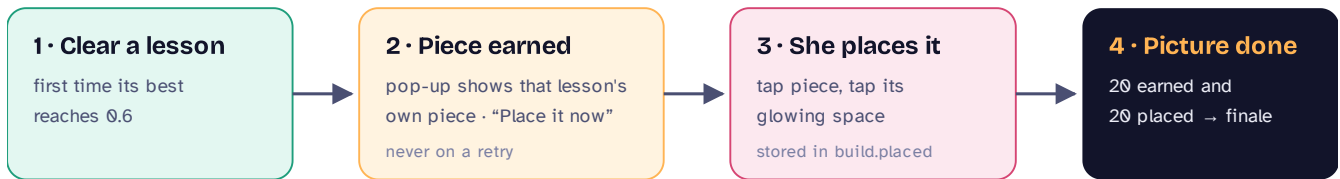
Questions (not activities) are shuffled once per attempt, so nothing can be memorised by position. Opening a hint never costs points; it only rules out mastery for that attempt.

Hands-on puzzle setting

MODE	TRY STEP	TEST STEP AND SCORE
Full	Shown.	The activity is graded by its check like any question.
Gentle	Shown.	Any real attempt counts as fully right (“That go counts”). An untouched activity still does not.
Skip	Skipped; the comic leads straight to Learn.	The activity is removed from the lesson and from the points available, so the lesson is scored on its questions alone.

Work from the Try step carries into the Test step: if she already did the activity, the Test shows her result with a “Have another go” option instead of a blank activity.

Earn, place, reveal



Waiting pieces (earned but not placed) show as a number on the Puzzle tab so none are forgotten.

```

piece for lesson L = its position in the course (world order, then lesson order)
earned(piece) = lesson L is cleared // derived
placed(piece) = in build.placed AND earned // unearned entries are ignored
waiting = earned and not placed
picture complete = all 20 earned AND all 20 placed
  
```

A save file that claims all 20 pieces are placed, with no lessons cleared, earns nothing, shows no pieces on the board and does not open the picture. This is covered by a test that plays the whole course: a half-finished lesson earns nothing, each cleared lesson earns exactly one piece, and the picture opens only at 20 of 20.

The jigsaw

- **Layout.** 4 columns × 5 rows = 20 pieces, over one hidden picture drawn as plain SVG at 400 × 500.
- **Edges.** Outer edges are flat. Each inner seam gets a tab on one side and a matching blank on the other, chosen by a fixed seed, so the layout is the same on every device.
- **Meshing.** Every tab uses one bump curve that is symmetric end to end, pushed outward along that edge's true direction; the neighbouring blank uses the same curve pushed inward. They trace the identical line, so there are no gaps. A test compares the two outlines point by point.
- **Hard to guess.** Waiting pieces are tilted between -25° and $+25^\circ$, and empty spaces show only a faint dashed outline.

What each check looks for

Activities report what she did as a small state object. A named check turns that into a score. Checks are forgiving on purpose: they reward thinking about each part, not exact wording.

ACTIVITY	STATE REPORTED	CHECK
Prompt lab	How many of the four parts her request covers. Each part is spotted by keywords (e.g. “for a six-year-old”, “list”, “under 50 words”, “as a table”); a number with enough context counts as a limit. Needs at least 3 words to count anything.	prompt-parts: at least 3 of 4
Spot the made-up answer	Correct judgements out of 8 (4 true, 4 invented).	spotted-fakes: at least 6
Temperature dial	Whether she asked at the steady end (< 0.34) and the wild end (> 0.66).	felt-temperature: both ends
Task sorter	Tasks placed in their best bin, out of 8.	sorted-tasks: at least 6
How hard to check	Answers matched to the right level of care, out of 6.	checked-care: at least 4
Pipeline builder	Steps in their right place, out of 4.	built-pipeline: all 4
Keep the agent honest	Wrong turns caught, out of 3 plans.	checked-agent: at least 2

Backup, import and settings

ACTION	RULE
Autosave	Every recorded attempt and setting change is written straight to localStorage.
Load	Migrated to the current version step by step; unreadable data starts a fresh profile rather than crashing.
Import a file	Must parse, migrate and validate, or it is refused and nothing changes. Records for lessons not in this build are dropped.
Pace switch	Switches between Explorer and Builder reading pace; never removes records.
PIN	Exactly four digits, or it is cleared. When set, the grown-ups page asks for it each time it is opened.
Start over	Two-step confirm, then an empty profile.